

Call Center Electrical Design

Mohamed Abdella



Presentation Aims & Goals

Lighting Design

Street Lighting

Small Power & Equipment loads

Earthing & Lightning

HVAC Load

VD & SC Using Excel Sheet

Panel Schedule

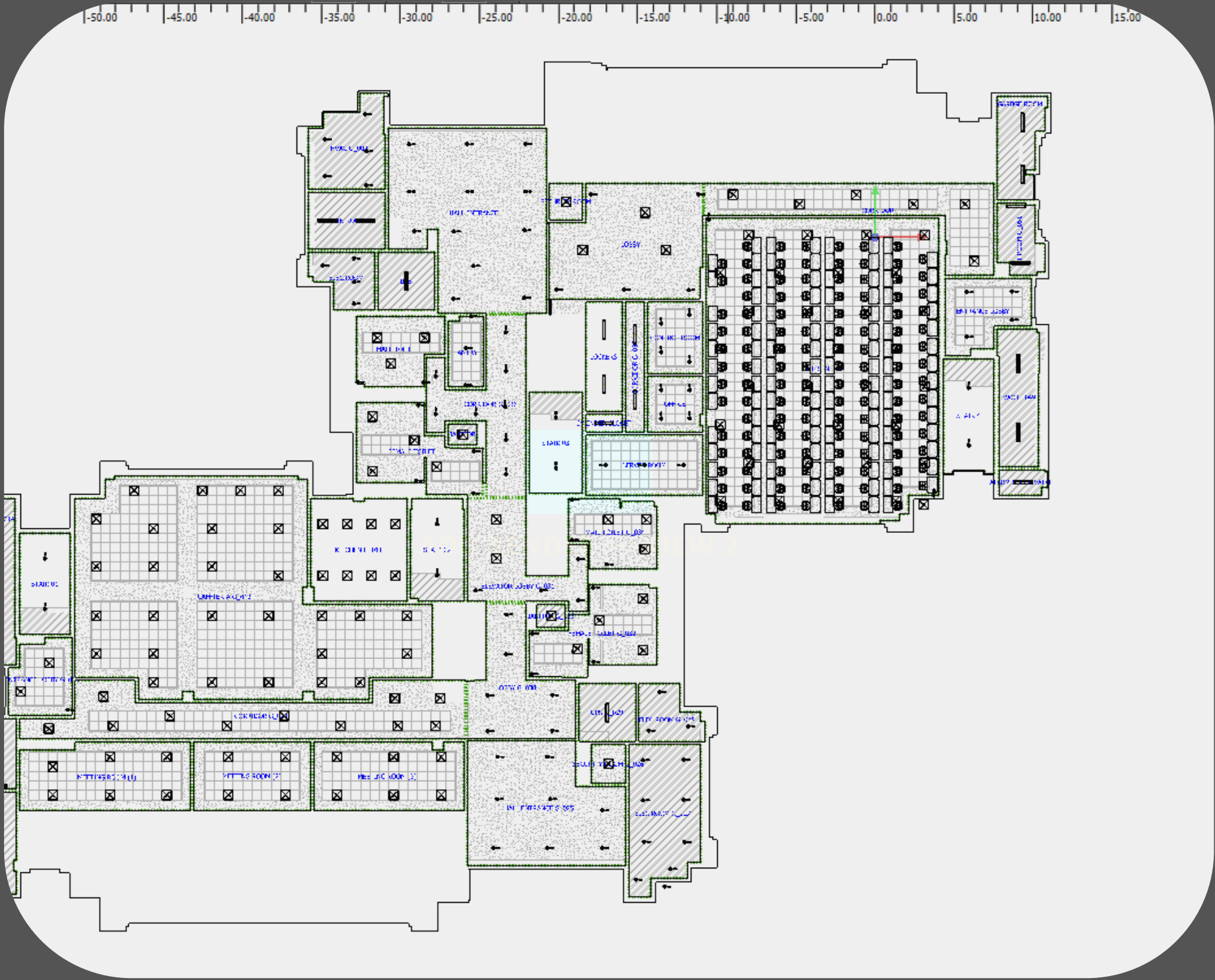
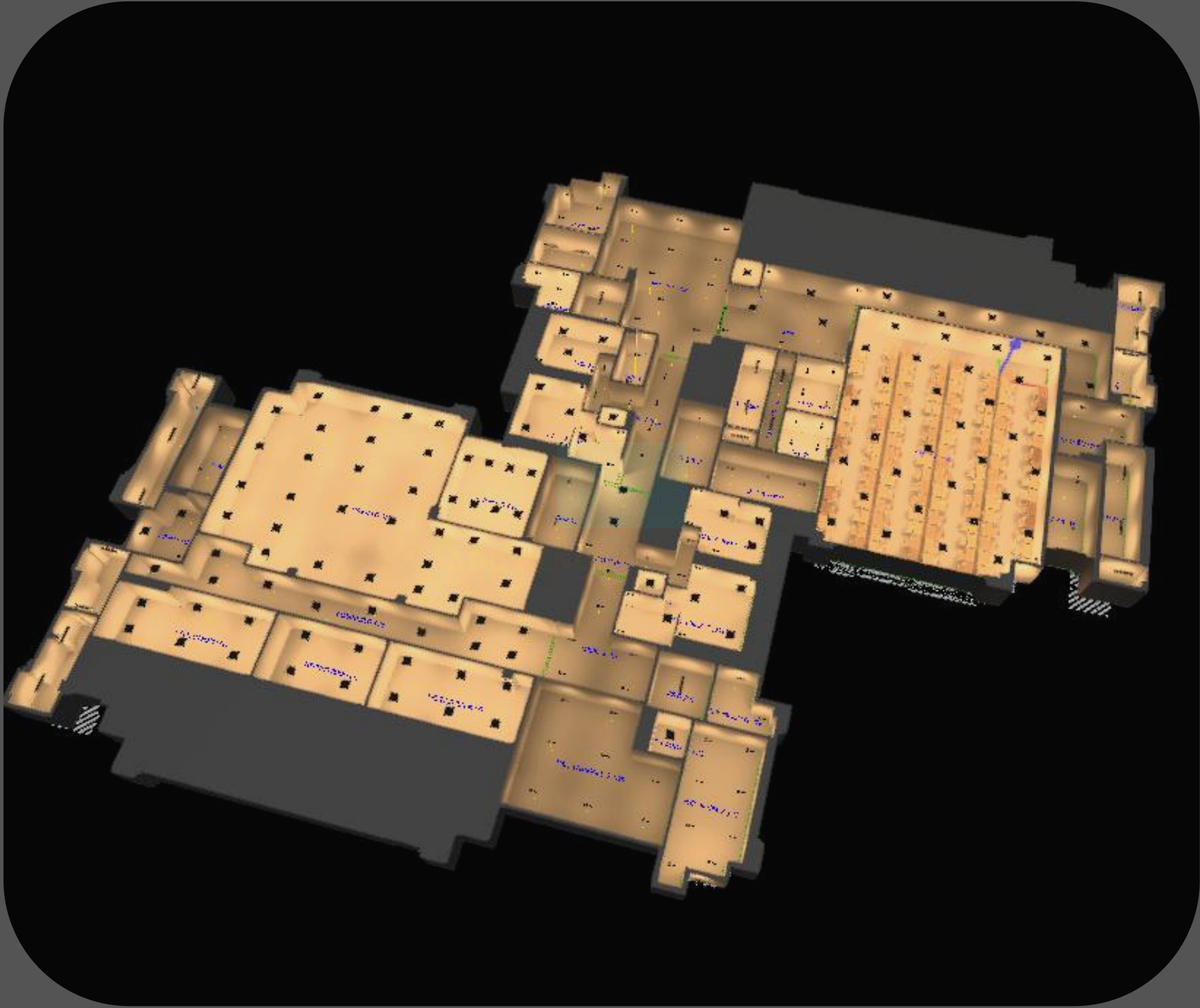
Load Estimation

Single Line Diagram

Bill OF Quantities

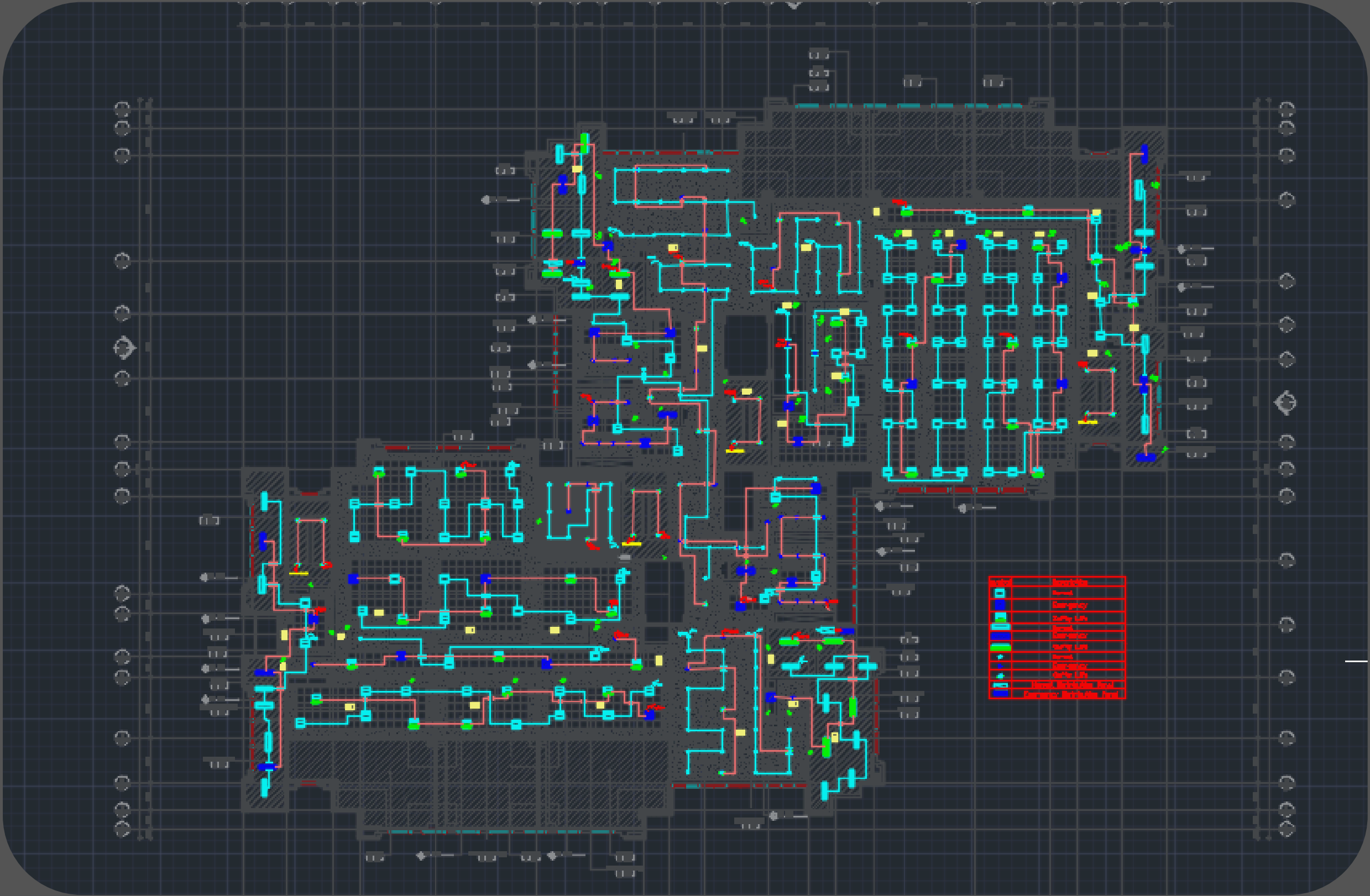
Lighting Design

DIALUX evo



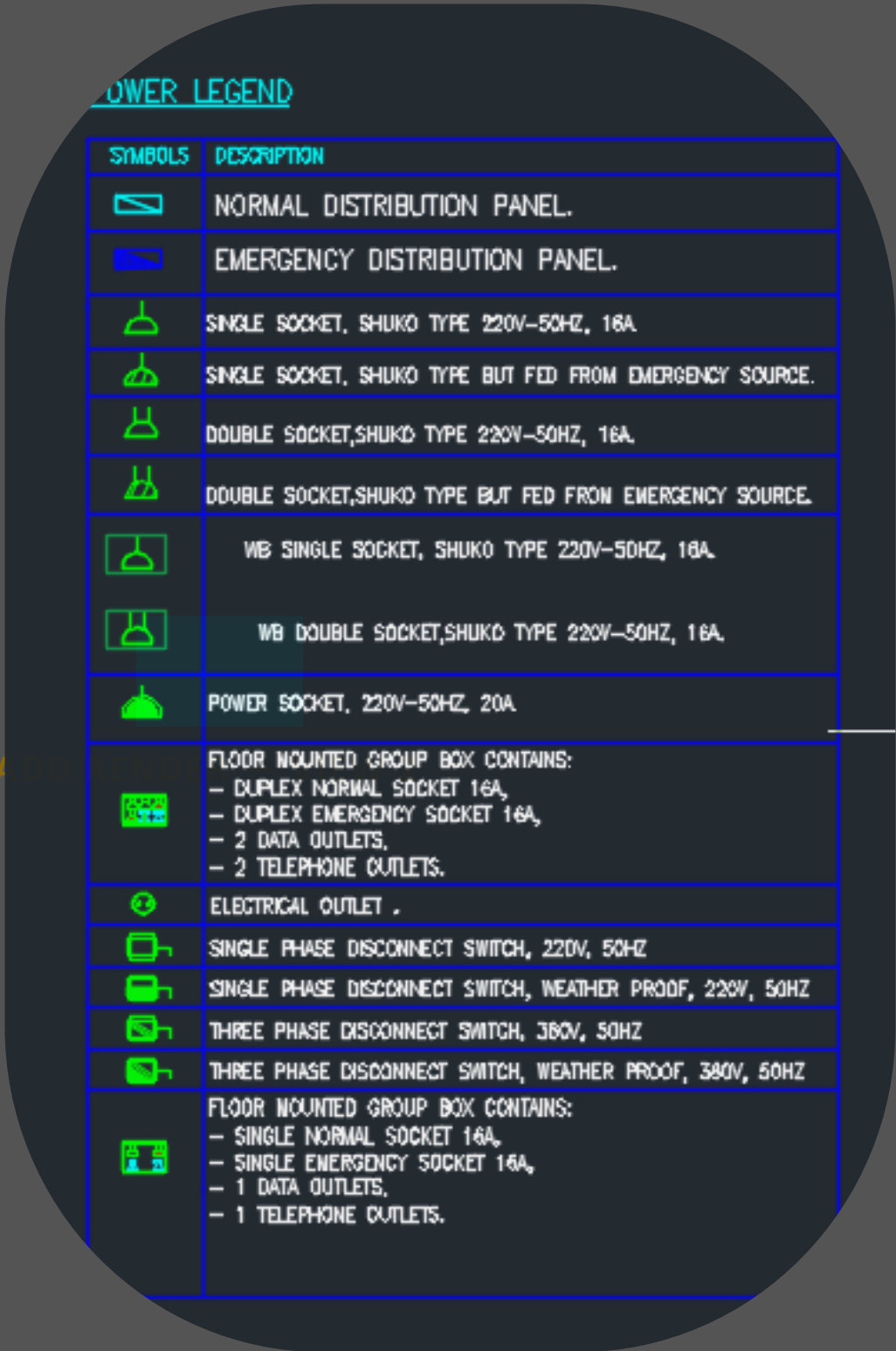
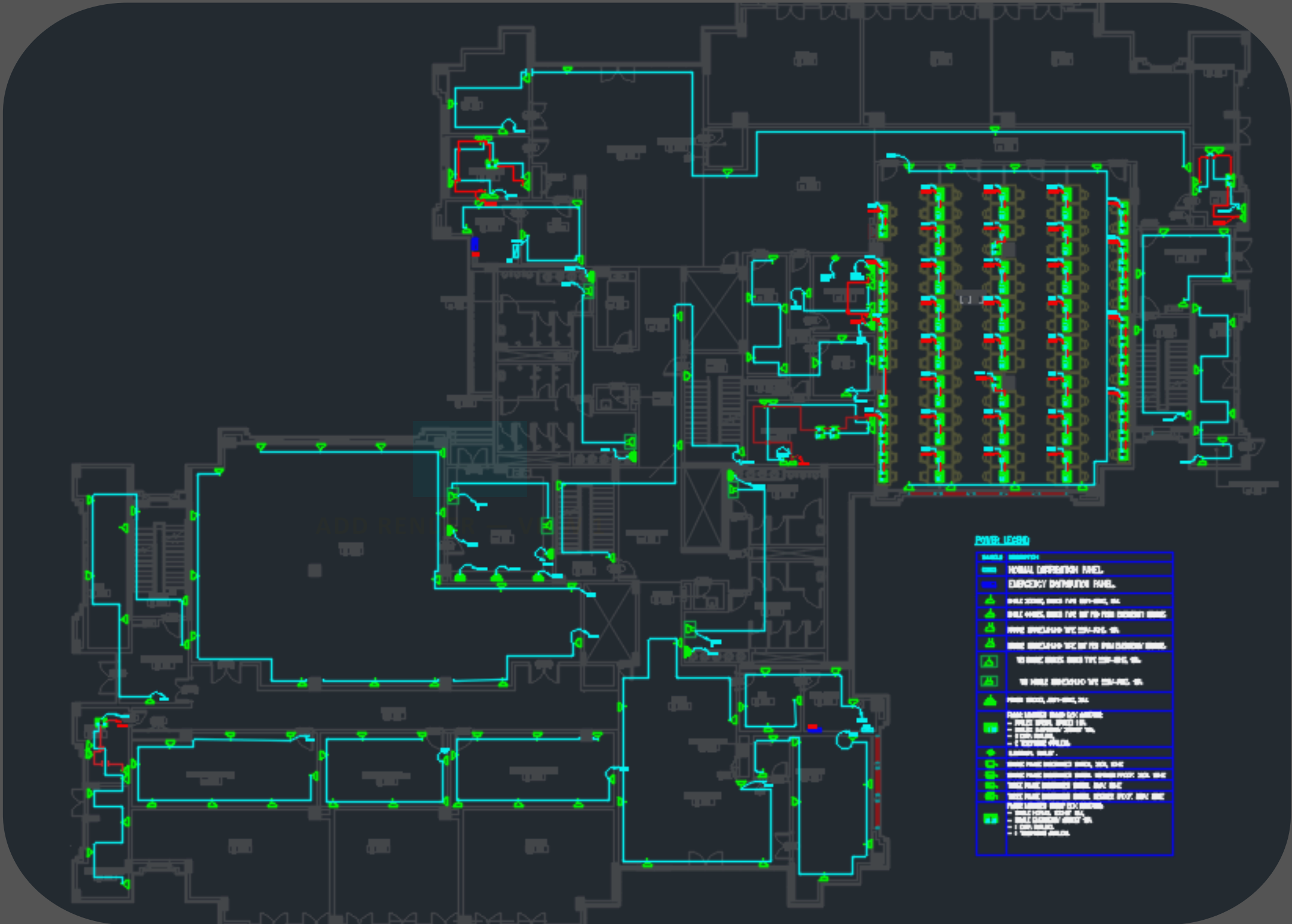
Lighting Design

Autocad



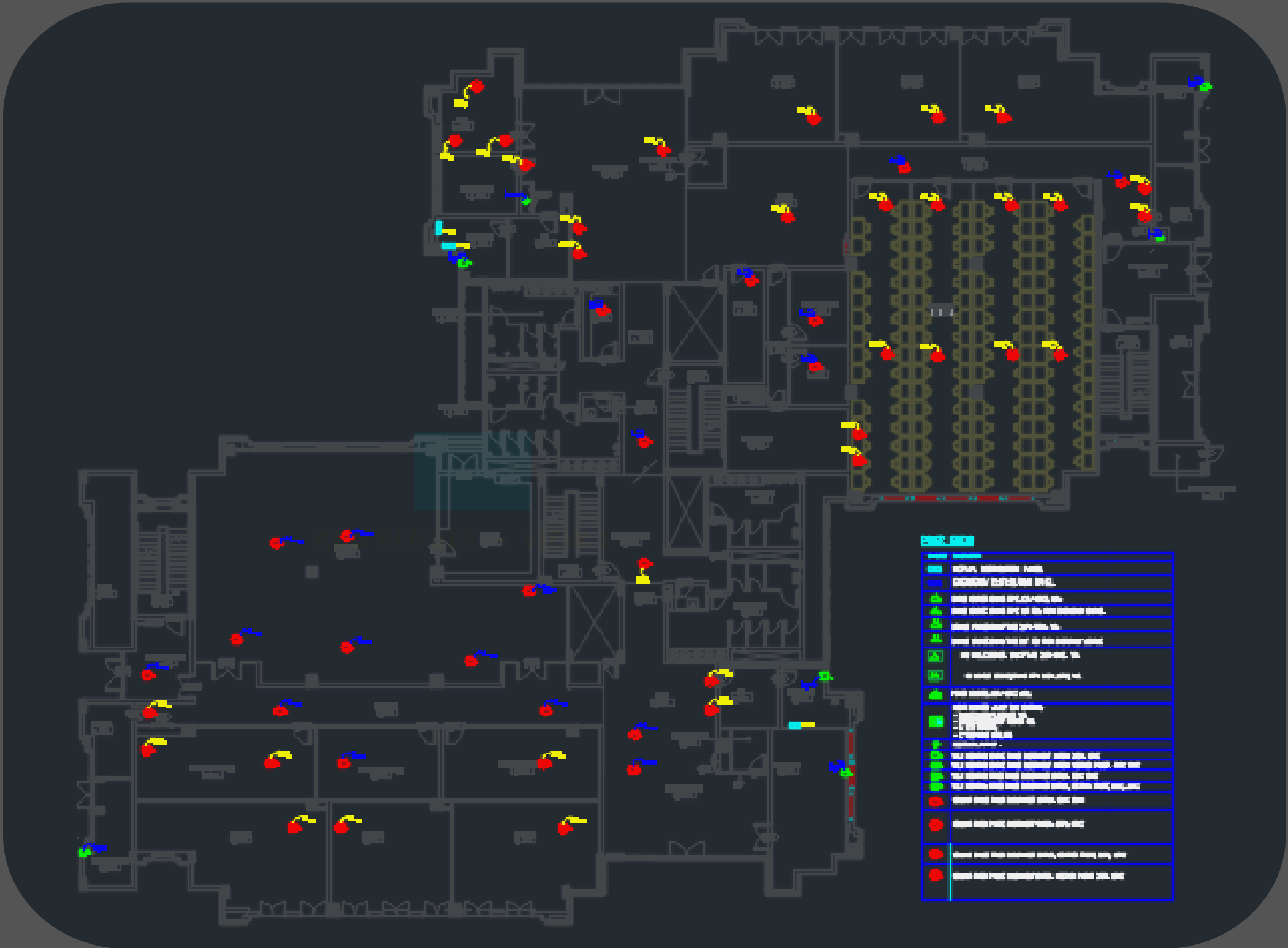
Small Power

Autocad



HVAC Load

Autocad



Panal Schedule

LIGHTING

DESIGN

SIZE OF CONDUIT (MM)	SIZE OF WIRE (MM2)	PANEL NAME : LP-G1										MOUNTED : SURFACE										400 / 230, 3 PHASE, 4W&G :										AMP. TRIP		/		AMP. FRAME		SIZE OF WIRE	SIZE OF CONDUIT
		LOCATION : BASEMENT										ROOM NO. :										MAIN CIRCUIT BREAKER RATING :										80		/		100			
		CT. NO.	DESCRIPTION	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	NO. OF PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS ⚡	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1			BR No	A	B	C	BR No	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2			POWER FACTOR COS ⚡	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO. OF PH	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	DESCRIPTION	CT. NO.							
												(VA)	(W)	(VAR)	No				No	(VAR)	(W)	(VA)		(AMP)	(AMP)	(DF2)		(VA)	(VA)	(VA)		No.	(MM2)	(MM)					
20 mm	3*3	1	Lighting	247.0			1	0.90	16		0.95	222	211	69	1	*			2	76	232	245	0.95		16	0.90	1	271.70			lighting	2	3*3	20 mm					
20 mm	3*3	3	Lighting		250.80		1	0.90	16		0.95	226	214	70	3	*			4	62	190	200	0.95		16	0.90	1		222.30		lighting	4	3*3	20 mm					
20 mm	3*3	5	Lighting			345.80	1	0.90	16		0.95	311	296	97	5	*			6	66	262	276	0.95		16	0.90	1			306.9	lighting	6	3*3	20 mm					
20 mm	3*3	7	Lighting	102.6			1	0.90	16		0.95	92	88	29	7	*			8	24	73	77	0.95		16	0.90	1	85.5			lighting	8	3*3	20 mm					
20 mm	3*3	9	Lighting		85.50		1	0.90	16		0.95	77	73	24	9	*			10	29	88	92	0.95		16	0.90	1		102.60		lighting	10	3*3	20 mm					
		11	spare				1	0.90	16		0.95	0	0	0	11	*			12	0	0	0					1				space	12							
SUB TOTAL VA				349	336	345						926	882	289						277	845	890						357	324	306	SUB TOTAL VA				BALANCE %				
TOTAL CONNECTED LOAD				=	2.02	KVA	DEMAND CURRENT "I" =				3	AMPERE	Total Active Power				=	2	KW	TAN ⚡ =				0.33	TOTAL CONNECTED LOAD 'A' =				706	VA	706	7.79							
DEMAND FACTOR "DF"				=	0.90		SHORT CIRCUIT LEVEL =				15	KA	Total Reactive Power				=	1	KVAR	COS ⚡ =				0.95	TOTAL CONNECTED LOAD 'B' =				660	VA	651								

SIZE OF CONDUIT (MM)	SIZE OF WIRE (MM2)	PANEL NAME : LP-G2										MOUNTED : SURFACE										400 / 230, 3 PHASE, 4W&G : AMP.TRIP / AMP. FRAME										SIZE OF WIRE (MM2)	SIZE OF CONDUIT (MM)	
		LOCATION : BASEMENT										ROOM NO. :										MAIN CIRCUIT BREAKER RATING : 80 / 100												
		CT. NO.	DESCRIPTION	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	NO. OF PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS ⚡	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1			BR No	A	B	C	BR No	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2			POWER FACTOR COS ⚡	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO. OF PH	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	DESCRIPTION	CT. NO.		
												(VA)	(W)	(VAR)	No				No	(VAR)	(W)	(VA)						(VA)	(VA)	(VA)			(MM2)	(MM)
20mm	3*3	1	Lighting	94.1			1	0.90	16		0.95	85	80	26	1	*			2	24	73	77	0.95		16	0.90	1	85.50			lighting	2	3*3	20mm
20mm	3*3	3	Lighting		212.80		1	0.90	16		0.95	192	182	60	3	*			4	40	122	128	0.95		16	0.90	1		142.50		lighting	4	3*3	20mm
20mm	3*3	5	Lighting			229.90	1	0.90	16		0.95	207	197	65	5	*			6	67	203	214	0.95		16	0.90	1			237.5	lighting	6	3*3	20mm
20mm	3*3	7	Lighting	237.5			1	0.90	16		0.95	214	203	67	7	*			8	19	58	62	0.95		16	0.90	1	68.4			lighting	8	3*3	20mm
20mm	3*3	9	Lighting		56.05		1	0.90	16		0.95	50	48	16	9	*			10	25	76	80	0.95		16	0.90	1		88.35		lighting	10	3*3	20mm
		11	spare				1	0.90	16		0.95	0	0	0	11	*			12	0	0	0					1				space	12		
SUB TOTAL VA				331	268	229						747	709	233						174	532	560					153	230	237	SUB TOTAL VA		MIN & MAX BALANCE %		
TOTAL CONNECTED LOAD				=	1.45	KVA	DEMAND CURRENT "I" = 2 AMPERE				Total Active Power = 1 KW				TAN ⚡ = 0.33				TOTAL CONNECTED LOAD "A" = 484 VA 498				6.43											
DEMAND FACTOR "DF"				=	0.90		SHORT CIRCUIT LEVEL = 15 KA				Total Reactive Power = 0 KVAR				COS ⚡ = 0.95				TOTAL CONNECTED LOAD "B" = 498 VA 466															

Panal Schedule

Small power

SIZE	SIZE	PANEL NAME : PP-G1										MOUNTED : SURFACE										400 / 230, 3 PHASE, 4W&G :										AMP. TRIP	/	AMP. FRAME	SIZE	SIZE
OF	OF	LOCATION : BASEMENT										ROOM NO. :										MAIN CIRCUIT BREAKER RATING :										90	/	100	OF	OF
CONDUIT	WIRE	CT.	DESCRIPTION	PHASE (A)	PHASE (B)	PHASE (C)	NO. OF PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS ϕ	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1			BR	A	B	C	BR	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2			POWER FACTOR COS ϕ	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO. OF PH	PHASE (A)	PHASE (B)	PHASE (C)	DESCRIPTION	CT.	WIRE	CONDUIT		
(MM)	(MM2)	NO.		(VA)	(VA)	(VA)		(DF1)				(VA)	(W)	(VAR)	No				No	(VAR)	(W)	(VA)		(AMP)	(AMP)	(DF2)	PH	(VA)	(VA)	(VA)		NO.	(MM2)	(MM)		
25 mm	3*3	1	SOCKTS	1750			1	0.80	20		0.80	1400	1120	840	1	*			2	840	1120	1400	0.80		20	0.80	1	1750				SOCKTS	2	3*3	25 mm	
25 mm	3*3	3	SOCKTS		1750		1	0.80	20		0.80	1400	1120	840	3	*			4	960	1280	1600	0.80		20	0.80	1		2000			SOCKTS	4	3*3	25 mm	
25 mm	3*3	5	SOCKTS			2000	1	0.80	20		0.80	1600	1280	960	5	*			6	960	1280	1600	0.80		20	0.80	1			2000			SOCKTS	6	3*3	25 mm
25 mm	3*3	7	SOCKTS	1750			1	0.80	20		0.80	1400	1120	840	7	*			8	840	1120	1400	0.80		20	0.80	1	1750				SOCKTS	8	3*3	25 mm	
25 mm	3*3	9	SOCKTS		2000		1	0.80	20		0.80	1600	1280	960	9	*			10	720	960	1200	0.80		20	0.80	1		1500			SOCKTS	10	3*3	25 mm	
25 mm	3*3	11	SOCKTS			1500	1	0.80	20		0.80	1200	960	720	11	*			12	720	960	1200	0.80		20	0.80	1			1500			SOCKTS	12	3*3	25 mm
25 mm	3*3	13	SOCKTS	1500			1	0.80	20		0.80	1200	960	720	13	*			14	720	960	1200	0.80		20	0.80	1	1500				SOCKTS	14	3*3	25 mm	
25 mm	3*3	15	SOCKTS		1500		1	0.80	20		0.80	1200	960	720	15	*			16	480	640	800	0.80		20	0.80	1		1000			SOCKTS	16	3*3	25 mm	
25 mm	3*3	17	SOCKTS			1500	1	0.80	20		0.80	1200	960	720	17	*			18	840	1120	1400	0.80		20	0.80	1			1750			SOCKTS	18	3*3	25 mm
25 mm	3*3	19	SOCKTS	1500			1	0.80	20		0.80	1200	960	720	19	*			20	840	1120	1400	0.80		20	0.80	1	1750				SOCKTS	20	3*3	25 mm	
25 mm	3*3	21	SOCKTS		2000		1	0.80	20		0.80	1600	1280	960	21	*			22	600	800	1000	0.80		20	0.80	1		1250			SOCKTS	22	3*3	25 mm	
25 mm	3*3	23	SOCKTS			1500	1	0.80	20		0.80	1200	960	720	23	*			24	720	960	1200	0.80		20	0.80	1			1500			SOCKTS	24	3*3	25 mm
25 mm	3*3	25	SOCKTS	1500			1	0.80	20		0.80	1200	960	720	25	*			26	600	800	1000	0.80		20	0.80	1	1250				SOCKTS	26	3*3	25 mm	
25 mm	3*3	27	SOCKTS		1750		1	0.80	20		0.80	1400	1120	840	27	*			28	480	640	800	0.80		20	0.80	1		1000			SOCKTS	28	3*3	25 mm	
25 mm	3*3	29	SOCKTS			2000	1	0.80	20		0.80	1600	1280	960	29	*			30	0	0	0	0.80		20	0.80	1					SPARE	30	3*3	25 mm	
25 mm	3*3	31	Spare				1	0.80	20		0.80	0	0	0	31	*			32	0	0	0	0.80		20	0.80	1					SPARE	32	3*3	25 mm	
25 mm	3*3	33	Spare				1	0.80	20		0.80	0	0	0	33	*			34	0	0	0					1					SPACE	34			
25 mm	3*3	35	Spare				1	0.80	20		0.80	0	0	0	35	*			36	0	0	0					1					SPACE	36			
25 mm	3*3	37	Spare				1	0.80	20		0.80	0	0	0	37	*			38	0	0	0					1					SPACE	38			
25 mm	3*3	39	Spare				1	0.80	20		0.80	0	0	0	39	*			40	0	0	0					1					SPACE	40			
25 mm	3*3	41	Spare				1	0.80	20		0.80	0	0	0	41	*			42	0	0	0					1					SPACE	42			
		SUB TOTAL VA		8000	9000	8500						20400	16320	12240						10320	13760	17200						8000	6750	6750	SUB TOTAL VA			BALANCE %		
		TOTAL CONNECTED LOAD		=	47.00	KVA		DEMAND CURRENT "I" =		54	AMPERE	Total Active Power		=	30	KW	TAN ϕ =		0.75								TOTAL CONNECTED LOAD 'A' =		16000	VA	16000	4.69				
		DEMAND FACTOR "DF"		=	0.80			SHORT CIRCUIT LEVEL =		15	KA	Total Reactive Power		=	23	KVAR	COS ϕ =		0.80								TOTAL CONNECTED LOAD 'B' =		15750	VA	15250					
		DEMAND LOAD		=	37.60	KVA											SIN ϕ =		0.60							TOTAL CONNECTED LOAD 'C' =		15250	VA							

Small power

EPE Grup & Mohamed Emam

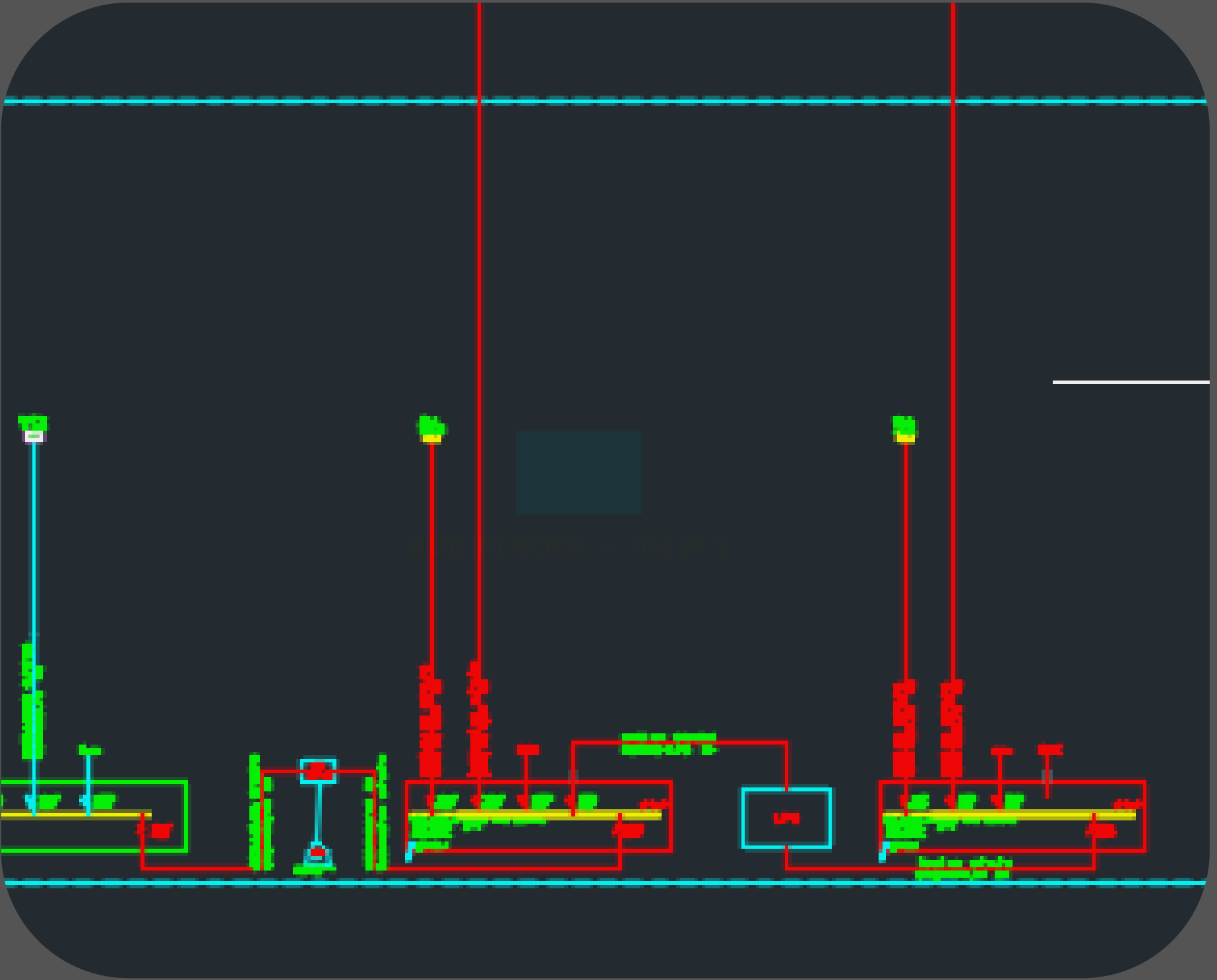
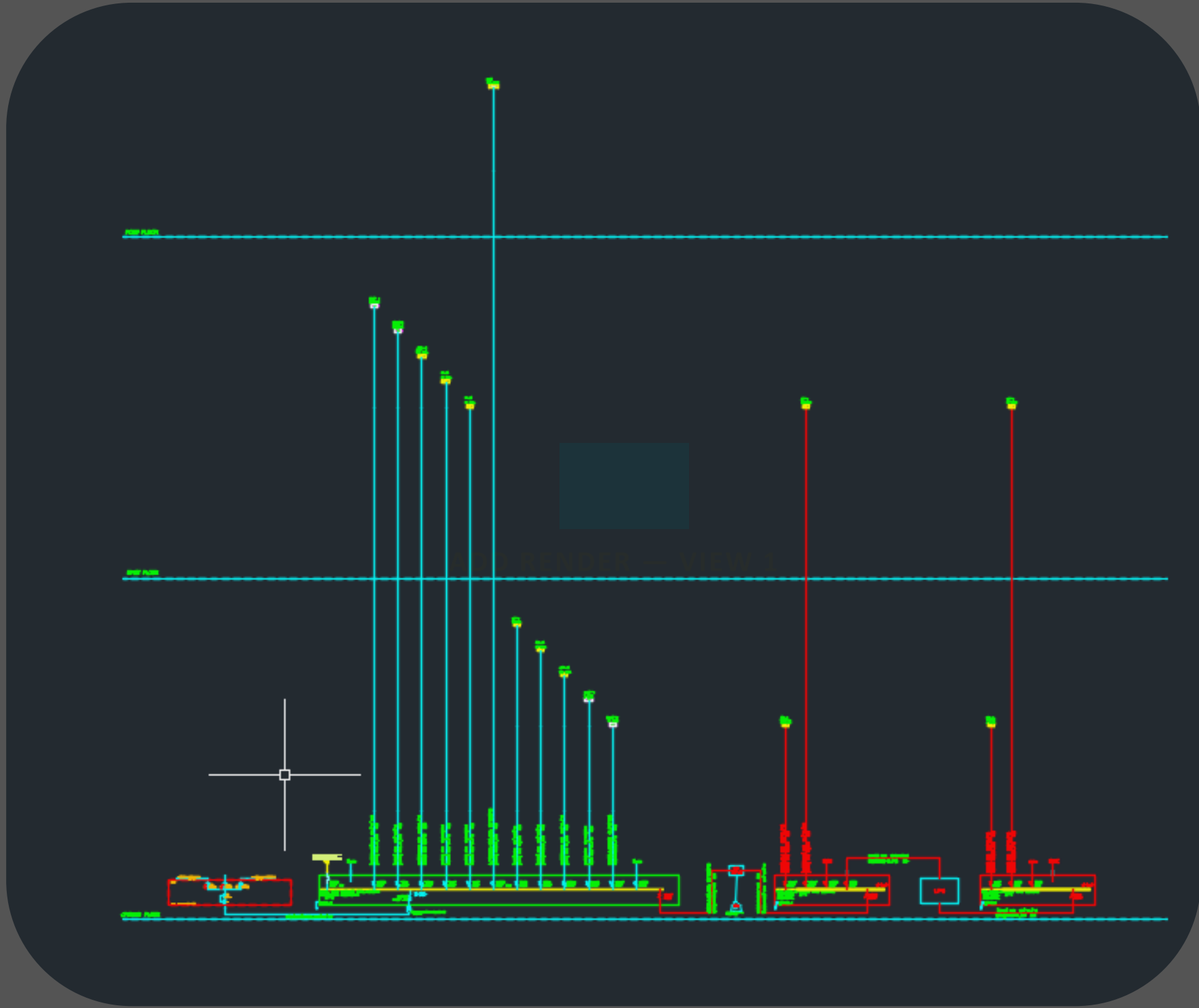
Panal Schedule

HVAC Load

SIZE OF	SIZE OF	PANEL NAME : LOCATION :		HVAC-G2 BASEMENT			MOUNTED : ROOM NO. :			SURFACE			400 / 230, 3 PHASE, 4W&G : MAIN CIRCUIT BREAKER RATING :										AMP.TRIP 90	I I	AMP. FRAME 100	SIZE OF	SIZE OF								
CONDUIT (MM)	WIRE (MM2)	CT. NO.	DESCRIPTION	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	NO. OF PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS ϕ	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1 (VA) (W) (VAR)			BR No	A	B	C	BR No	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2 (VAR) (W) (VA)			POWER FACTOR COS ϕ	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO. OF PH	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	DESCRIPTION	CT. NO.	WIRE (MM2)	CONDUIT (MM)	
25 mm	4*4+1*4	1	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	1	*			2	900	1200	1500	0.80		25	0.80	1	1875					2	4*4+1*4	25 mm
		3			1875		1	0.80	25		0.80	1500	1200	900	3	*			4	900	1200	1500	0.80		25	0.80	1		1875				4		
		5				1875	1	0.80	25		0.80	1500	1200	900	5	*			6	900	1200	1500	0.80		25	0.80	1			1875			6		
25 mm	4*4+1*4	7	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	7	*			8	900	1200	1500	0.80		25	0.80	1	1875					8	4*4+1*4	25 mm
		9			1875		1	0.80	25		0.80	1500	1200	900	9	*			10	900	1200	1500	0.80		25	0.80	1		1875				10		
		11				1875	1	0.80	25		0.80	1500	1200	900	11	*			12	900	1200	1500	0.80		25	0.80	1			1875			12		
25 mm	4*4+1*4	13	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	13	*			14	900	1200	1500	0.80		25	0.80	1	1875					14	4*4+1*4	25 mm
		15			1875		1	0.80	25		0.80	1500	1200	900	15	*			16	900	1200	1500	0.80		25	0.80	1		1875				16		
		17				1875	1	0.80	25		0.80	1500	1200	900	17	*			18	900	1200	1500	0.80		25	0.80	1			1875			18		
25 mm	4*4+1*4	19	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	19	*			20	900	1200	1500	0.80		25	0.80	1	1875					20	4*4+1*4	25 mm
		21			1875		1	0.80	25		0.80	1500	1200	900	21	*			22	900	1200	1500	0.80		25	0.80	1		1875				22		
		23				1875	1	0.80	25		0.80	1500	1200	900	23	*			24	900	1200	1500	0.80		25	0.80	1			1875			24		
25 mm	4*4+1*4	25	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	25	*			26	900	1200	1500	0.80		25	0.80	1	1875					26	4*4+1*4	25 mm
		27			1875		1	0.80	25		0.80	1500	1200	900	27	*			28	900	1200	1500	0.80		25	0.80	1		1875				28		
		29				1875	1	0.80	25		0.80	1500	1200	900	29	*			30	900	1200	1500	0.80		25	0.80	1			1875			30		
25 mm	4*4+1*4	31	HVAC 3PH	1875			1	0.80	25		0.80	1500	1200	900	31	*			32	1200	1600	2000	0.80		32	0.80	1	2500					32	3*10	20 mm
		33			1875		1	0.80	25		0.80	1500	1200	900	33	*			34	1200	1600	2000	0.80		32	0.80	1		2500				34	3*10	20 mm
		35				1875	1	0.80	25		0.80	1500	1200	900	35	*			36	1200	1600	2000	0.80		32	0.80	1			2500			36	3*10	20 mm
20 mm	3*10	37	Spare				1	0.80	25		0.80	0	0	0	37	*			38	1200	1600	2000	0.80		32	0.80	1	2500					38	3*10	20 mm
20 mm	3*10	39	HVAC 1PH		2500		1	0.80	32		0.80	2000	1600	1200	39	*			40	720	960	1200	0.80		16	0.80	1		1500				40	3*10	20 mm
20 mm	3*10	41	HVAC 1PH			2500	1	0.80	32		0.80	2000	1600	1200	41	*			42	0	0	0	0.80		25	0.80	1					42	3*10	20 mm	
20 mm	3*10	43	SPARE				1	0.80	25		0.80				43	*			44	0	0	0	0.80		25	0.80	1					44			
20 mm	3*10	45	SPARE				1	0.80	25		0.80				45	*			46	0	0	0	0.80		25	0.80	1					46			
20 mm	3*10	47	SPARE				1	0.80	25		0.80				47	*			48	0	0	0	0.80		25		1					48			
		SUB TOTAL VA		11250	13750	13750						31000	24800	18800					19020	25380	31700						14375	13375	11875	SUB TOTAL VA			BALANCE %		
TOTAL CONNECTED LOAD		=	78.38	KVA	DEMAND CURRENT "I" =		91	AMPERE	Total Active Power		=	50	KW	TAN ϕ =		0.75	TOTAL CONNECTED LOAD 'A' =		25625	VA	27125	5.53													
DEMAND FACTOR "DF"		=	0.80	SHORT CIRCUIT LEVEL =		15	KA	Total Reactive Power		=	38	KVAR	COS ϕ =		0.80	TOTAL CONNECTED LOAD 'B' =		27125	VA	25625															
DEMAND LOAD		=	62.70	KVA									SIN ϕ =		0.60	TOTAL CONNECTED LOAD 'C' =		25625	VA																

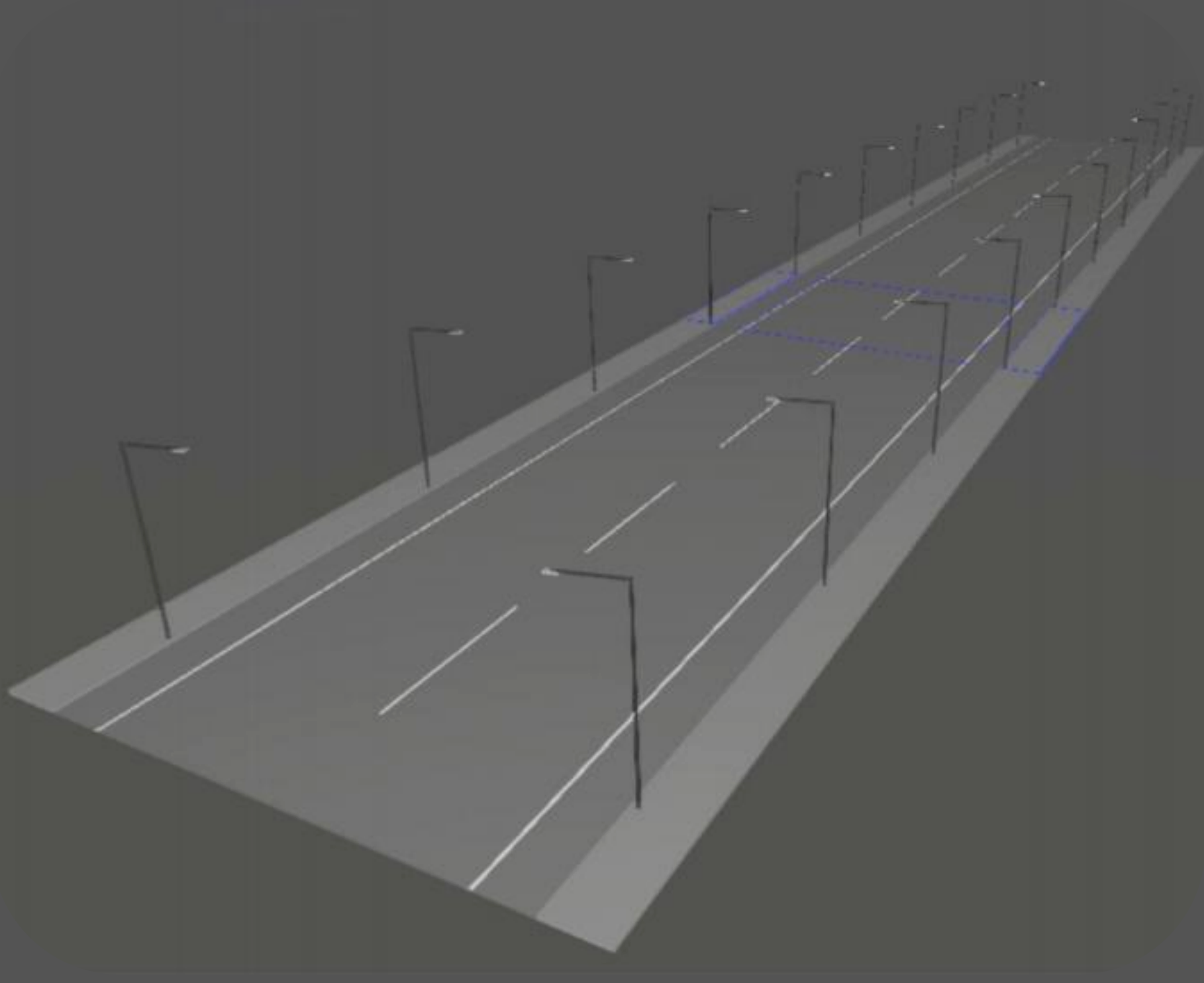
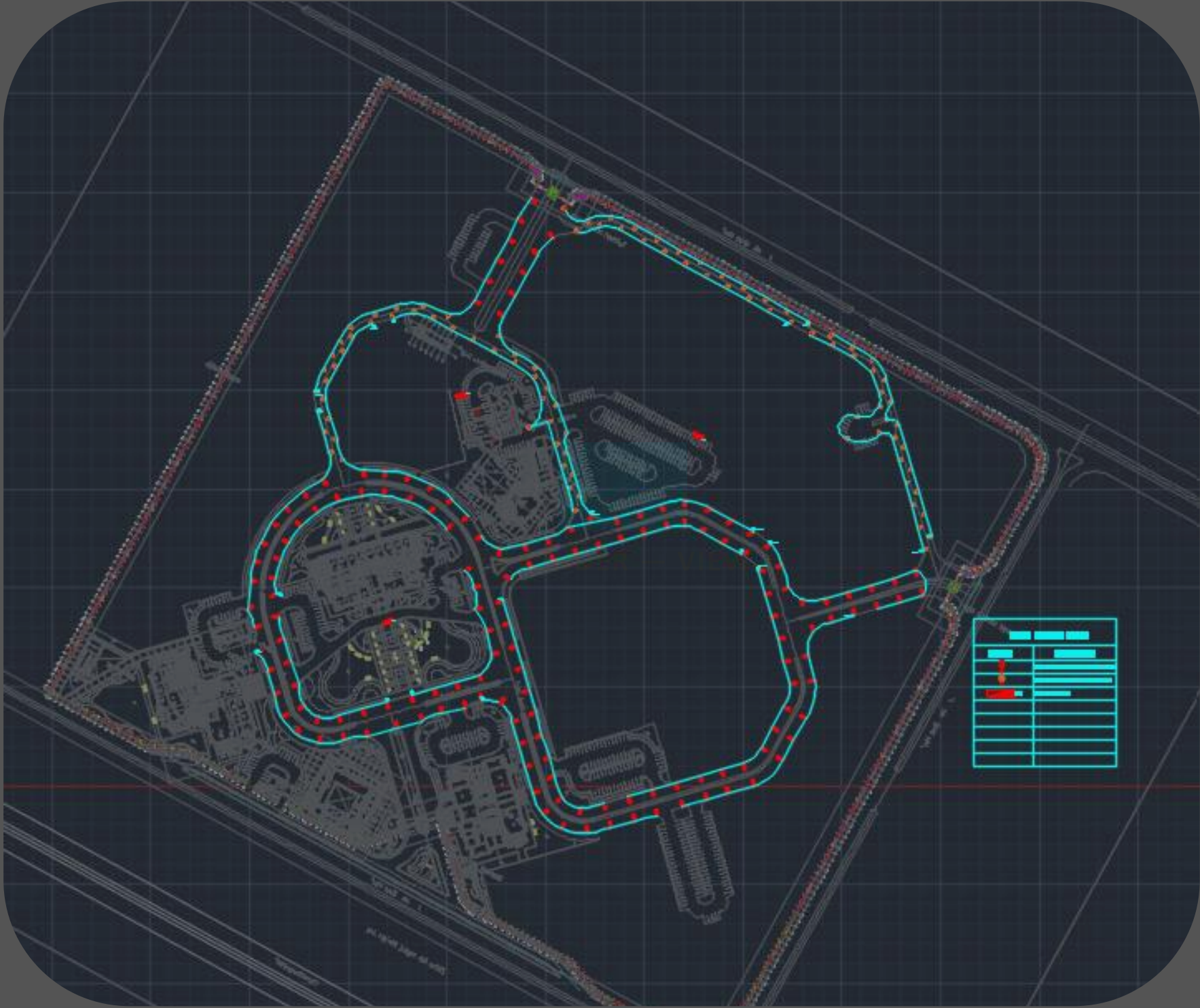
Single Line Diagram

Autocad



Street Lighting

Autocad



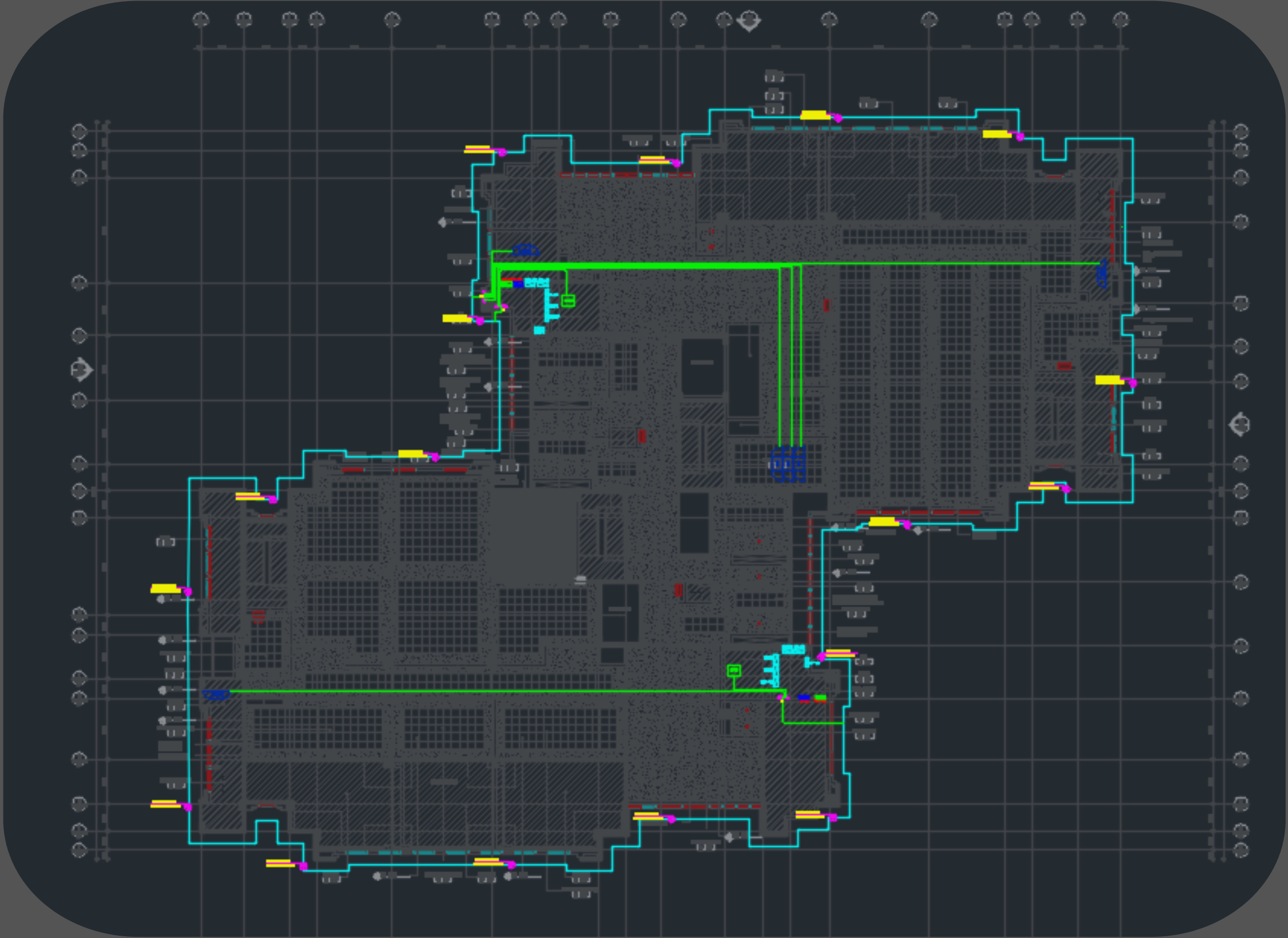
Panal Schedule

SIZE OF CONDUIT	SIZE OF WIRE	PANEL NAME : LOCATION :	SL-P1 Site						400 / 230, 3 PHASE, 4W&G : MAIN CIRCUIT BREAKER RATING :	AMP. TRIP 90	/ 100	AMP. FRAME 100		SIZE OF WIRE	SIZE OF CONDUIT																			
CONDUIT (MM)	WIRE (MM2)	CT. NO.	DESCRIPTION	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	NO. OF PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS Φ	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1 (VA) (W) (VAR)	BR No	A	B	C	BR No	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2 (VA) (W) (VAR)	POWER FACTOR COS Φ	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO. OF PH	PHASE (A) (VA)	PHASE (B) (VA)	PHASE (C) (VA)	DESCRIPTION	CT. NO.	WIRE (MM2)	CONDUIT (MM)				
		1		190				1.00	32		0.95	190 181 59	1 *				2	47 144 152	0.95		32	1.00		152				2						
under ground	4*16	3	Street Lighting		190		3	1.00	32		0.95	190 181 59	3 *				4	47 144 152	0.95		32	1.00	3		152		Street Lighting	4	4*16	under ground				
		5				190		1.00	32		0.95	190 181 59	5 *				6	47 144 152	0.95		32	1.00				152			6					
under ground	4*16	7	Street Lighting	152				1.00	32		0.95	152 144 47	7 *				8	47 144 152	0.95		32	1.00		152				8						
		9	Street Lighting		152		3	1.00	32		0.95	152 144 47	9 *				10	47 144 152	0.95		32	1.00	3		152		Street Lighting	10	4*16	under ground				
		11				152		1.00	32		0.95	152 144 47	11 *				12	47 144 152	0.95		32	1.00				152			12					
under ground	4*16	13	Street Lighting	224			3	1.00	32		0.95	224 213 70	13 *				14	87 266 280	0.95		32	1.00		280			Street Lighting	14	4*16	under ground				
		15			224			1.00	32		0.95	224 213 70	15 *				16	87 266 280	0.95		32	1.00	3		280			16	4*16	under ground				
		17				224		1.00	32		0.95	224 213 70	17 *				18	87 266 280	0.95		32	1.00				280			18					
under ground	4*16	19	Street Lighting	224			3	1.00	32		0.95	224 213 70	19 *				20	70 213 224	0.95		32	1.00		224			Street Lighting	20	4*16	under ground				
		21			224			1.00	32		0.95	224 213 70	21 *				22	70 213 224	0.95		32	1.00	3		224			22	4*16	under ground				
		23				224		1.00	32		0.95	224 213 70	23 *				24	70 213 224	0.95		32	1.00				224			24					
under ground	4*16	25	Street Lighting	224			3	1.00	32		0.95	224 213 70	25 *				26	71 215 228	0.95		32	1.00		226			Street Lighting	26	4*16	under ground				
		27			224			1.00	32		0.95	224 213 70	27 *				28	71 215 228	0.95		32	1.00	3		226			28	4*16	under ground				
		29				224		1.00	32		0.95	224 213 70	29 *				30	71 215 228	0.95		32	1.00				226			30					
under ground	4*16	31	Spare				1	1.00	32		0.95	0 0 0	31 *				32	0 0 0	0.95		32	1.00	1				spare	32	4*16	under ground				
under ground	4*16	33	Spare				1	1.00	32		0.95	0 0 0	33 *				34	0 0 0	0.95		32	1.00	1				Space	34						
under ground	4*16	35	Spare				1	1.00	32		0.95	0 0 0	35 *				36	0 0 0	0.95		32	1.00	1				Space	36						
		SUB TOTAL VA		1014	1014	1014						3042 2689 949					968 2946 3102							1034 1034 1034			SUB TOTAL VA			BALANCE %				
TOTAL CONNECTED LOAD				=	6.14	KVA		DEMAND CURRENT "I" =				9	AMPERE	Total Active Power				=	6	KW	TAN Φ =				0.33	TOTAL CONNECTED LOAD "A" =				2048	VA	2048	0.00	
DEMAND FACTOR "DF"				=	1.00			SHORT CIRCUIT LEVEL =				15	KA	Total Reactive Power				=	2	KVAR	COS Φ =				0.95	TOTAL CONNECTED LOAD "B" =				2048	VA	2048		
DEMAND LOAD				=	6.14	KVA																SIN Φ =				0.31	TOTAL CONNECTED LOAD "C" =				2048	VA		

PANEL NAME :		SL-P2				MOUNTED :				SURFACE				400 / 230, 3 PHASE, 4W&G :				AMP.TRIP		I	AMP. FRAME		SIZE		SIZE							
LOCATION :		Site				ROOM NO. :				out door				MAIN CIRCUIT BREAKER RATING :				90		I	100		OF		OF							
DESCRIPTION		PHASE (A)	PHASE (B)	PHASE (C)	NO.O F PH	DEMAND FACTOR (DF1)	AMP TRIP (AMP)	AMP FRAME (AMP)	POWER FACTOR COS ϕ	INDIVIDUAL DEMAND LOAD (A+B+C)*DF1			BR	A	B	C	BR	INDIVIDUAL DEMAND LOAD (A+B+C)*DF2			POWER FACTOR COS ϕ	AMP FRAME (AMP)	AMP TRIP (AMP)	DEMAND FACTOR (DF2)	NO.O F PH	PHASE (A)	PHASE (B)	PHASE (C)	DESCRIPTION	CT.	WIRE	CONDUIT
		(VA)	(VA)	(VA)						(VA)	(W)	(VAR)	No				No	(VAR)	(W)	(VA)						(VA)	(VA)	(VA)		NO.	(MM2)	(MM)
Street Lighting		168			3	1.00	32		0.95	168	160	52	1	*			2	52	160	168	0.95		32	1.00		168				2		
			168			1.00	32		0.95	168	160	52	3	*			4	52	160	168	0.95		32	1.00			168			4	4*16	under ground
			168			1.00	32		0.95	168	160	52	5	*			6	52	160	168	0.95		32	1.00				168			6	
Street Lighting		224			3	1.00	32		0.95	224	213	70	7	*			8	70	213	224	0.95		32	1.00		224				8		
			224			1.00	32		0.95	224	213	70	9	*			10	70	213	224	0.95		32	1.00			224			10	4*16	under ground
			224			1.00	32		0.95	224	213	70	11	*			12	70	213	224	0.95		32	1.00				224			12	
Street Lighting		224			3	1.00	32		0.95	224	213	70	13	*			14	70	213	224	0.95		32	1.00		224				14		
			224			1.00	32		0.95	224	213	70	15	*			16	70	213	224	0.95		32	1.00			224			16	4*16	under ground
			224			1.00	32		0.95	224	213	70	17	*			18	70	213	224	0.95		32	1.00				224			18	
SPARE					1	1.00	32		0.95	0	0	0	19	*			20	0	0	0	0.95		32	1.00	1				SPARE	20	4*16	under ground
SPARE					1	1.00	32		0.95	0	0	0	21	*			22	0	0	0	0.95		32	1.00	1				space	22		
SPARE					1	1.00	32		0.95	0	0	0	23	*			24	0	0	0	0.95		32	1.00	1				space	24		
TOTAL VA		616	616	616						1848	1755	577					577	1755	1848						616	616	616	SUB TOTAL VA			BALANCE %	
LOAD	=	3.70	KVA		DEMAND CURRENT "I" =				5	AMPERE	Total Active Power				=	4	KW	TAN ϕ =				0.33	TOTAL CONNECTED LOAD 'A' =				1232	VA	1232	0.00		
DF"	=	1.00			SHORT CIRCUIT LEVEL =				15	KA	Total Reactive Power				=	1	KVAR	COS ϕ =				0.95	TOTAL CONNECTED LOAD 'B' =				1232	VA	1232			
	=	3.70	KVA															SIN ϕ =				0.31	TOTAL CONNECTED LOAD 'C' =				1232	VA				

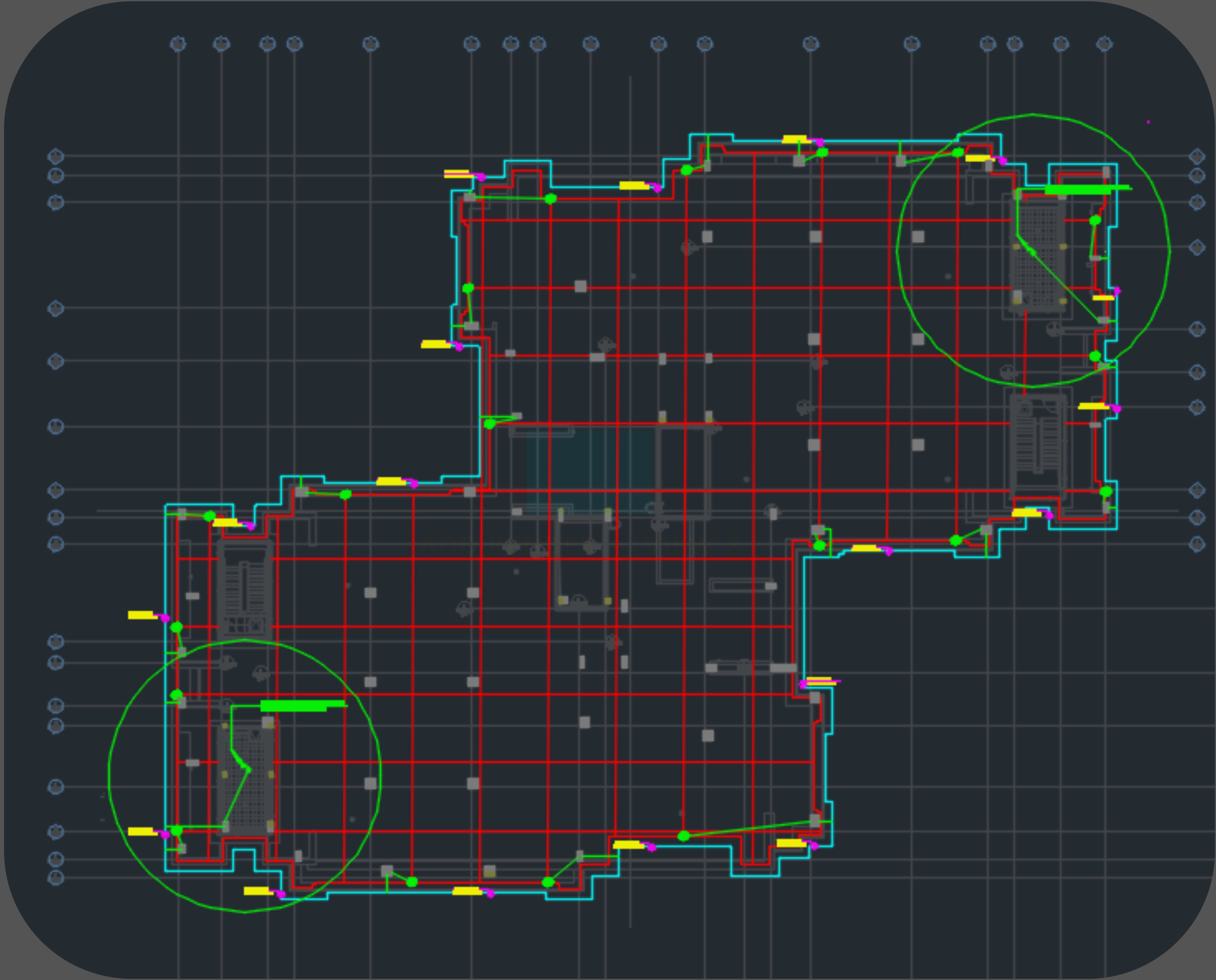
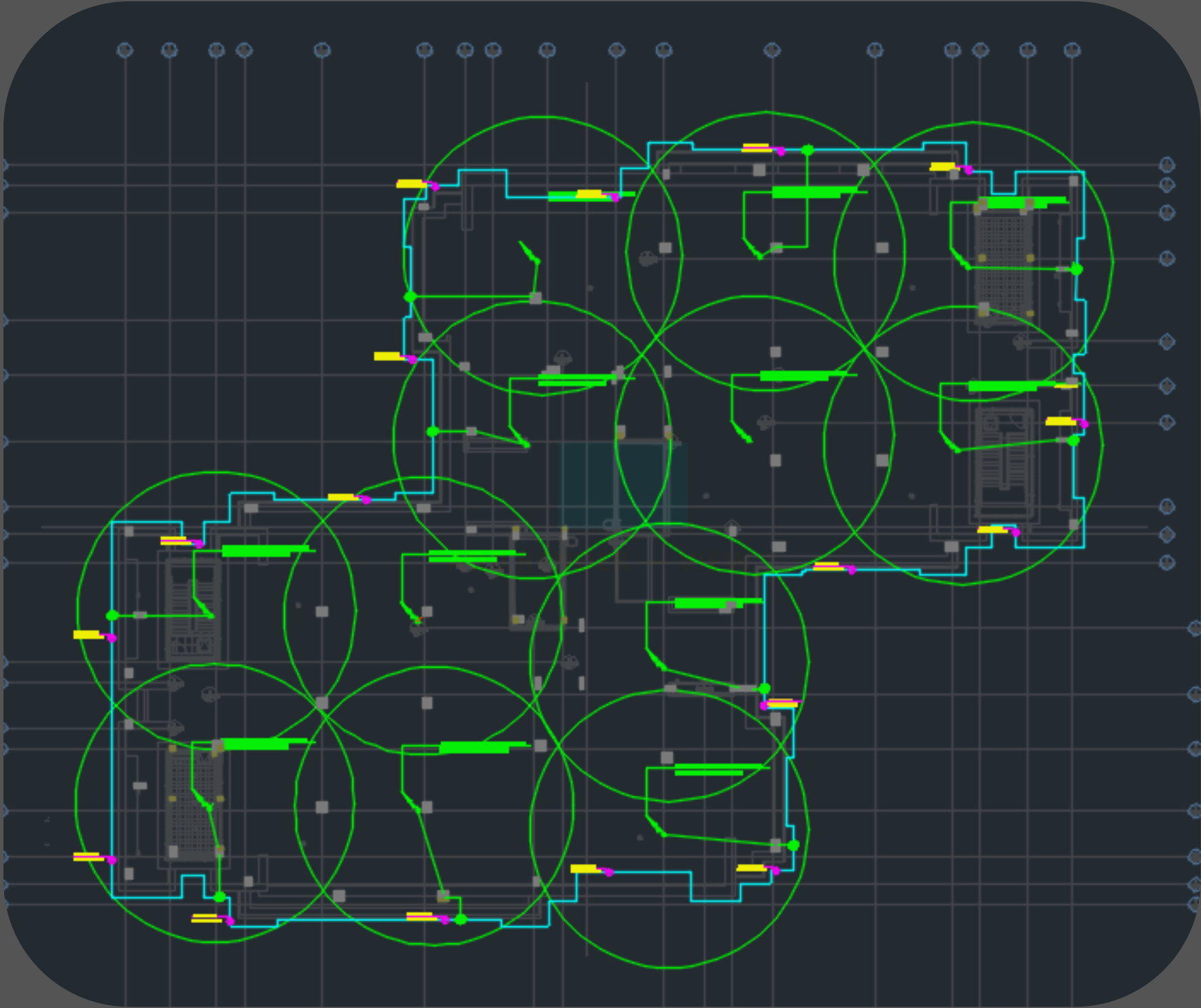
Earthing

Autocad



Lightning

Autocad



Voltage Drop & Short Circuit

Excel

				SC Calculation Sheet							
Calculated With Eng. :				Mohamed Abdella					Date : 27/8/2026		
From	TO Panel	V rated(KV)	Z%	TR. Rating (MVA)	Z (TR.)	Cable	mV/A/m	LENGTH	Z-Cable	I.s.c. (KA)	Rated(KA)
Transformer	MDB	0.22	0.06	1	0.00871	(3C*300+1CX150mm2)AL/XLPE/STA/PVC	0.255	100	0.0147	9.4	10
MDB	MCC	0.22	0.06	1	0.00871	(3CX120+1CX70)mm2 CU/PVC/PVC	0.287	30	0.0050	16.1	18
MDB	LP-G	0.22	0.06	1	0.00871	(4CX10)mm2 CU/PVC/PVC	3.101	20	0.0358	4.9	6
MDB	PP-G	0.22	0.06	1	0.00871	(4CX10)mm2 CU/PVC/PVC	3.101	20	0.0358	4.9	6
MDB	ACP-G	0.22	0.06	1	0.00871	(3CX70+1x35)mm2 CU/PVC/PVC	0.402	20	0.0046	16.5	18
MDB	SHOP(1)	0.22	0.06	1	0.00871	(4CX10)mm2 CU/PVC/PVC	3.101	70	0.1253	1.6	4.5
MDB	SHOP(2)	0.22	0.06	1	0.00871	(4CX25)mm2 CU/PVC/PVC	0.957	55	0.0304	5.6	6
MDB	LP-G	0.22	0.06	1	0.00871	(4CX4)mm2 CU/PVC/PVC	7.741	10	0.0447	4.1	4.5
MDB	PP-G	0.22	0.06	1	0.00871	(4CX10)mm2 CU/PVC/PVC	3.101	10	0.0179	8.3	10
MDB	ACP-G	0.22	0.06	1	0.00871	(3CX70+1x35)mm2 CU/PVC/PVC	0.402	10	0.0023	19.9	25
MDB	SHOP(1)	0.22	0.06	1	0.00871	(3CX4)mm2 CU/PVC/PVC	7.741	60	0.2682	0.8	4.5
MDB	SHOP(2)	0.22	0.06	1	0.00871	(4CX25)mm2 CU/PVC/PVC	0.957	45	0.0249	6.6	10

		VD Calculation Sheet										
Calculated With Eng. :		Mohamed Abdella					Date : 27/8/2026					
FROM PANEL	To Panel	Length(m)	Load(KVA)	Current(A)	(mV/A/m)	Cable	No Of Cables	V.D(volt)	V.D %	TOTAL V.D. %		
										FROM	TO	
Transformer	MDB	100	450	681.8181818	0.255	(3C*300+1CX150mm2)AL/XLPE/STA/PVC	7	2.483766234	0.62%	Transformer	MDB	0.62%
MDB	MCC	30	300	454.5454545	0.287	(3CX120+1CX70)mm2 CU/PVC/PVC	4	0.978409091	0.24%	MDB	MCC	0.66%
MDB	LP-G	20	15	22.72727273	3.101	(4CX10)mm2 CU/PVC/PVC	1	1.409545455	0.35%	MDB	LP-G	0.77%
MDB	PP-G	20	18	27.27272727	3.101	(4CX10)mm2 CU/PVC/PVC	1	1.691454545	0.42%	MDB	PP-G	0.84%
MDB	ACP-G	20	65	98.48484848	0.402	(3CX70+1x35)mm2 CU/PVC/PVC	1	0.791818182	0.20%	MDB	ACP-G	0.62%
MDB	SHOP(1)	70	15	22.72727273	3.101	(4CX10)mm2 CU/PVC/PVC	1	4.933409091	1.23%	MDB	SHOP(1)	1.65%
MDB	SHOP(2)	55	30	45.45454545	0.957	(4CX25)mm2 CU/PVC/PVC	1	2.3925	0.60%	MDB	SHOP(2)	1.02%
MDB	LP-G	10	6	9.090909091	7.741	(4CX4)mm2 CU/PVC/PVC	1	0.703727273	0.18%	MDB	LP-G	0.60%
MDB	PP-G	10	17	25.75757576	3.101	(4CX10)mm2 CU/PVC/PVC	1	0.798742424	0.20%	MDB	PP-G	0.62%
MDB	ACP-G	10	62	93.93939394	0.402	(3CX70+1x35)mm2 CU/PVC/PVC	1	0.377636364	0.09%	MDB	ACP-G	0.51%
MDB	SHOP(1)	60	4	6.060606061	7.741	(3CX4)mm2 CU/PVC/PVC	1	2.814909091	0.70%	MDB	SHOP(1)	1.12%
MDB	SHOP(2)	45	30	45.45454545	0.957	(4CX25)mm2 CU/PVC/PVC	1	1.9575	0.49%	MDB	SHOP(2)	0.91%

Load Estimation by IEC & SBC

Excel

Electrical Load Estimation by SBC					
Floor	Area	Estimated	Connected Load	Diversity Factor	Demand Load
Ground Floor	2150	0.22	473	0.7	331
First Floor	2500	0.22	550	0.7	385
Roof	2500	0.05	125	0.8	100
Shops	350	0.22	77	0.7	54
TDL					870
CL (Largest)			1148		
C1 (Shops)			77		
CDL	838				

Electrical load Estimation by IEC			
Floor	Area	Estimated	KVA
Ground Floor	2500	0.12	300
First Floor	2500	0.12	300
Roof	2500	0.04	100
Hvac			500
Lifts			9
TL			1209
Diversity Fac			0.8
TDL			967.2

		كابات الجهد المنخفض و الموصلات	4
		بالمتر الطولي لتوريد وتركيب واختبار وتشغيل كابلات الجهد المنخفض متعددة الأقطاب 600/1000 فولت . بالقطاعات التالية والسعر يشمل تركيب واختبار الكابل (والترهيات محملة في بند منفصل) وكل ما يلزم لإتمام العمل طبقا لأصول الصناعة وكما هو موضح بالرسومات والمواصفات وطرق القياس الخاصة بالمشروع كالاتي :	
700	م	(3C*300+1CX150mm2)AL/XLPE/STA/PVC	4.1
120	م	(3CX120+1CX70)mm2 CU/PVC/PVC	4.2
30	م	(3CX70+1x35)mm2 CU/PVC/PVC	4.3
50	م	(4CX10)mm2 CU/PVC/PVC	4.4
10	م	(4CX4)mm2 CU/PVC/PVC	4.5
		توريد وتركيب واختبار وتشغيل كابلات الجهد المنخفض احادية القطب 450/750 بالقطاعات التالية والسعر يشمل تركيب واختبار الكابل (والترهيات محملة في بند منفصل) وكل ما يلزم لإتمام العمل طبقا لأصول الصناعة وكما هو موضح بالرسومات والمواصفات وطرق القياس الخاصة بالمشروع كالاتي :	5
300	م	1C*300mm2 ECC AL/XLPE/PVC	5.1
120	م	1CX70mm2 CU/PVC ECC	5.2
30	م	1X35mm2 CU/PVC ECC	5.3
50	م	1X10mm2 CU/PVC ECC	5.4
10	م	1X4mm2 CU/PVC ECC	5.5

		الانارة	7
		توريد وتركيب واختبار وتشغيل وحدات الانارة والسعر يشمل جسم الكشف والكابح والمبات وكل ما يلزم لاتمام العمل على الوجه الاكمل وذلك طبقا لاصول الصناعة وما هو موضح بالرسومات والمواصفات الخاصة بالمشروع كالاتي :	
114	عدد	كشف انارة نموذج 60*60 توريد و تركيب و اختيار وحدة إضاءة ليد عبارة عن كشف 60*60 سم بقدره 45 وات ، بدرجة حماية 4000K:3200lm (IP23) من النوع الغاطس والفئة تشمل كل ما يلزم للتوريد والتركيب والإعتماد من الإستشاري ، و كل ما يلزم مما جميعه حسب المواصفات الفنية و أصول الصناعة .	7.1
29	عدد	كشف انارة نموذج 60*60 كابلند السابق ولكن ببطارية من النيكل كادميوم تكفي للتشغيل 3 ساعات بعد انقطاع التيار الكهربى	7.2
122	عدد	كشف انارة نموذج سيوت توريد و تركيب واختيار وحدة إضاءة ليد عبارة سيوت لايت بلمبة ليد قدره 6 وات بدرجة حماية (IP 23) من النوع الغاطس والفئة تشمل كل ما يلزم للتوريد والتركيب والاعتماد من الاستشاري و كل ما يلزم لإنهاء الاعمال حسب المواصفات الفنية و أصول الصناعة.	7.3
34	عدد	كشف انارة نموذج سيوت كابلند السابق ولكن ببطارية من النيكل كادميوم تكفي للتشغيل 3 ساعات بعد انقطاع التيار الكهربى	7.4
34	عدد	كشف انارة نموذج 120 سم توريد و تركيب و اختيار وحدة إضاءة ليد عبارة عن كشف 120 سم بقدره 42 وات ، 4000K من النوع السطحي (Surface mounted) والفئة تشمل كل ما يلزم للتوريد والتركيب والإعتماد من الإستشاري ، و كل ما يلزم لإنهاء الاعمال حسب المواصفات الفنية و أصول الصناعة.	7.5
8	عدد	كشف انارة نموذج 120 سم كابلند السابق ولكن ببطارية من النيكل كادميوم تكفي للتشغيل 3 ساعات بعد انقطاع التيار الكهربى	7.6

THANK YOU

Mohamed Waleed Abdella

◦ Electrical Design